Phase 3F | First-coach prototype integration

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Coordinate the first coach before cutting steel

PF-TRAIN-S3F-R0 | 18 October 2026 UTC

Preliminary engineering development - not for vehicle fabrication, construction, procurement release or passenger operation. This issue develops interfaces, installation space and a controlled build sequence. No supplier acceptance, physical test, measured mass or professional approval is claimed.

Phase 3E identified why the floor, wheels, boarding and frame cannot be selected independently. Phase 3F carries the raised-floor arrangement forward as the named **F1 integration study**, while keeping the original alternatives available. A 32-inch finished floor is a calculation and drawing input, not a released height. The larger Phase 3E P3 motion-envelope comparison still has a possible overlap at this height; this integration study does not resolve it.

New work	Practical use
Installation study	Locate service zones and circulation interfaces against the same frame, wheels, platforms and datum system.
Boarding and connections	Compare deployable boarding space, accessible interfaces and car articulation; identify incompatible arrangements before fabrication.
Mass and utilities	Trace allocated mass and centre of gravity, traction and brake demands, heating/cooling duty and electrical architecture.
Controlled first article	Define the order of specialist design, material release, fixture checks, installation, measurements and commissioning.
Review package	Eight coordinated drawing sheets, editable calculations, rotating installation study and blank fabrication/interface records.

01 | Controlled continuation and retained decisions

The saved Phase 3E report was read and its current identity verified on 18 October 2026. The technical site was confirmed at published version 7. Earlier reports, drawings, models and unresolved findings are preserved. This is a new Phase 3F revision, not a claim that previous hold points have been closed.

Retained basis	Meaning in this issue
Owner scope	36-inch evaluation gauge; diesel locomotive; three regular coaches plus one separate accessible coach; enclosed heating and cooling; hydraulic service-brake preference.
Site and operation	0% longitudinal grade is the target. North Pole location is deferred and passengers disembark there. Surveyed geometry, crosslevel, drainage and accepted construction tolerances remain necessary.
Cabin and occupancy	306 x 104 in clear cabin; 312 x 110 in body; 400 in frame; 32 regular places; 11 fixed accessible-coach seats and two wheelchair places. These remain proposed capacities pending circulation and vehicle approval.
Mass basis	7,040 / 7,535 kg regular / accessible tare allocations; 54,755 kg nominal, 59,840 kg stress and 65,100 kg upper train studies. 13,200 / 11,000 kg upper individual-car study cases are not supplier ratings.
Named study F1	Parent E1 frame, 32 in finished floor, all 110 retained steel member envelopes. No silent replacement of the 28 in predecessor or acceptance of the unresolved larger movement envelope.
Public edition	Technical material for Ralph, suppliers, fabricators and engineers. Business projections and attraction scenes are outside this issue. No outreach or orders performed.

02 | What can proceed and what closes the design

Proceed now with the ground-level interior and boarding-footprint layout, record actual spaces and equipment labels, and assemble the evidence required by the technical packets. Keep participants on stable existing ground with suitable rated chairs. Tape, templates and unsupported visual markers do not establish an occupied structure or vehicle rating.

Work package	Responsible completion before the following step
Geometry freeze	Vehicle and running-gear engineers resolve configured bogie/wheel/brake envelopes, floor height, suspension motion, stability, coupling movement and boarding interfaces.
Design and procurement release	Responsible specialists issue materials, connection details, welding/inspection basis, component ratings, system schematics and accepted design/test limits for the exact configuration.
First frame and body	Qualified fabrication follows the released drawings and recorded fixture/inspection sequence. Measure mass, dimensions and interfaces; document deviations before concealment.
Equipment installation	Specialists install brake, control, electrical, HVAC and boarding systems using coordinated approved interfaces. Farm crew assists only within specifically released work packages.
Unoccupied qualification	Complete staged static, fault, single-coach and complete-train verification using controlled test loads and accepted protocols. A successful file check does not substitute for these tests.
Duplication and operation	Resolve first-article findings, update the controlled configuration, and obtain required approvals and operating acceptance before copying coaches or carrying guests.

The package identifies actual catalog facts, analyst inputs, calculated demands, space reservations and held decisions separately. Empty actual-rating, tolerance, inspection and signoff fields mean the evidence has not been received. They must not be interpreted as zero or as acceptance.

Current county pathway

Clark County's signed Ordinance 5378 replaced Chapter 22.16, effective June 3, 2026. Section 22.16.020 addresses installations lasting at least 30 days, subject to exemptions and other jurisdiction. Twenty operating evenings do not by themselves establish an exemption. The Building Official determines classification under 22.16.120. [FJ01]

If this chapter applies, 22.16.130 requires Nevada design-professional coordination/review; 22.16.140 identifies sealed design documentation, site plan and operations/maintenance material. Sections 22.16.160-190 address fabrication quality records, installation inspection, an approved on-site Qualified Individual and county commissioning. Section 22.16.200 requires annual Certificate of Operation renewal. These are project planning inputs, not a classification or approval already received.

Appendix A lists ASTM F3054, F2291, F1193, F770, F2137, F3214 and F3598, among other standards. Its list does not print their edition years; 22.16.040/080 refers to Board-adopted editions. The principal designer must obtain the applicable edition matrix and resolve vehicle, station, accessibility and fire/electrical boundaries before release. The older permit guide remains historical process context, not proof of current requirements.

The final site plan still needs the North Pole location. Continue vehicle interface studies without inventing that alignment.

03 | Mass, structure and controlled fabrication

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F1 carries the E1 raised-floor arrangement forward for integration: 32 in finished floor, 31.003937 in steel bearing plane, 400 in frame and 216 in bogie spacing. Every one of the 110 E1 members is preserved. This phase adds installation reservations, a reconciled mass/CG scenario and controlled fabrication records; it does not release material, joints, holes, supports or tolerances. The 28 in floor remains the project baseline outside this named study.

Mass and vertical location

Catalog weights multiplied by retained member lengths give **1,527.288 kg bare steel**, with modeled CG at x153, y0, z26.500624 in. Atlas section-table entries were rechecked [C3FS01]. This excludes the required draft structure, joint reinforcement, brackets and attachments. Catalog mass is not an as-built weighing.

F1 planning bucket	Mass, kg	Evidence and treatment
Retained bare frame	1,527.288	Member-coordinate/catalog takeoff
Two bogie mechanical allowances	1,800	900 each; actual configuration unknown
Brakes and controls	250	Separate allocation; reconcile OEM package boundaries
Body, roof, glass and doors	900	575.303 partial takeoff; 324.697 unresolved
Cabin floor, insulation and finishes	400	146.789 partial takeoff; 253.211 unresolved
Sixteen bench sections	240	Structural study assumption: 15 each
Platform decks, stairs and guards	400	Excludes frame steel already counted
HVAC, ventilation and ducts	550	Retained allocation, not selected equipment mass
Electrical, batteries and lighting	240	Allocation
Couplings/intercar hardware	180	Allocation
Unassigned growth balance	552.712	Must fund unresolved details and overruns
Retained complete tare target	**7,040**	**Unmeasured target**

The original 1,500 kg frame and 180 kg seat allowances remain visible in the basis. Bare-frame excess consumes 27.288 kg of the original 640 kg growth allowance; the 240 kg bench study consumes another 60 kg. Unassigned growth is bookkeeping for eventual components, not ballast or available savings. Only 2,249.380 kg has a member/panel/conditional-glass takeoff; none has been weighed. Do not add a purchased braked bogie and the complete separate brake allowance without reconciling their contents.

The editable 19-row ledger assigns explicit low/reference/high vertical positions while retaining 7,040 kg. Reference examples are bogies z16 in, benches z44, electrical z24 and the 550 kg HVAC allocation at the generic roof-box center z133.614 in. Body/roof/floor rows have separate positions. The resulting empty-car CG is **z42.440 in reference**, versus 24.158 and 57.311 in simultaneous low/high position scenarios. These are not probability limits, feasible installation proofs or measured CG. Actual masses and coordinates remain blank.

Moving the same illustrative 550 kg from z133.614 to z24 lowers total CG by 217.5 mm at unchanged mass. That demonstrates the sensitivity; it does not show that a suitable HVAC package fits underfloor. A four-inch floor raise increases whole-car CG by the raised-mass fraction times 101.6 mm: fractions 0.5/0.75/1 give 50.8/76.2/101.6 mm. Actual sprung/unsprung allocation remains unresolved.

Occupancy and wheel-load sensitivity

Seat-row stations remain 41-265 in at 32 in pitch, with adult-center trials y±22/±42 in. Seated body CG is assumed 28 in above floor; one crew member is at x153, y0 and 42 in above floor. Stress adults are 145 kg. All mass, including both bogies, appears once in the gross reactions.

Case	Gross kg	Reference CG z, in	Front/rear resultant, kN
Empty allocation	7,040	42.440	34.519 / 34.519
32 nominal guests + crew, 100 kg each	10,340	48.179	50.700 / 50.700
32 stress guests + crew	11,825	49.717	57.982 / 57.982
16 stress guests on one side + crew	9,505	47.207	46.606 / 46.606
16 stress guests in front four rows + crew	9,505	47.207	53.347 / 39.865
Eight stress guests, front/one side + crew	8,345	45.429	44.289 / 37.548
Upper development screen	13,200	50.788	64.724 / 64.724

The upper case adds an explicit 1,375 kg residual at x153/y0/z60 to the full stress scenario. This is a load sensitivity, not permission to increase payload. Front/rear results are longitudinal wheel-contact resultants at the bogie-center planes, **not body-support forces**. Bogie self-weight and the actual sprung/unsprung split must be treated separately when designing body supports.

For actual contact separation b, total side reactions are W/2 ± M/b, where M=Σmgy+Σma_yz. Nominal 36 in track gauge does not establish b. The calculations therefore vary b=0.8/0.9144/1.0 m independently. Front roll-moment shares 0/0.5/1 and axle fractions 0.5/0.75 expose the missing stiffness/load-sharing information; they are not calibrated suspension predictions. Lateral accelerations 0/0.0382/0.1 g are sensitivities, not operating limits.

At b0.9144 m with equal front/rear roll and equal axle sharing, the one-side case gives illustrative wheel reactions 6.596-16.707 kN statically, or 3.540-19.763 kN at 0.1 g. An extreme sampled load-sharing assumption produces a negative reaction; the assumed no-uplift solution then cannot exist. Positive trial reactions do not establish derailment resistance. Actual profile/contact locations, springs, track twist/crosslevel, suspension roll, wind and transient crowding require a vehicle-dynamics assessment. Level longitudinal track does not remove these questions.

Installation reservations and load paths

The central proposed reservation is x93-213, y−30-30, z18-25 in. Its rectangular volume does not intersect the retained nominal frame members. This limited check says nothing about a complete bogie, brake hoses, deflection, service tools or removal space. The mechanical module’s retained P3 sampled wheel-proxy union ends at x82.721/223.279, leaving 10.279 in longitudinal separation from the reservation at either end; that proxy omits actual bogie hardware.

Systems boxes use only this reserved corridor. Direct side extraction at their installed elevation would cross the main girders, whose underside is z19.003937. A downward/pit or other designed removal route must clear the entire box below that underside plus an engineer-assigned margin before lateral movement. Actual clearance, support and handling arrangements remain blank.

The roof HVAC reservation is x88-218, y±35, z121.614-145.614 in; outside ducts are separate systems reservations. Neither roof bows nor mounting pads are selected. Equipment reactions must pass through designed supports into body posts/frame, not sandwich skins alone. Per 100 kg, each 1 g acceleration contributes 0.981 kN; a 300 mm offset adds 0.294 kNm. Required accelerations, combinations, bracket distribution, fatigue and fastener capacities are unassigned. All structural members are no-drill/no-cut/no-mount until a released detail identifies the modification.

Release, fabrication and inspection

The material decision precedes cutting. ASTM’s public A500 scope flags dynamic applications where notch toughness can matter [C3FS02]; a section table cannot settle fatigue or minimum-temperature suitability. Review alternatives and certified material properties with the vehicle/welding engineer [C3FS06]. AWS lists D15.1/D15.1M:2025 for railroad equipment, and D1.1/D1.1M:2025-AMD1 for structural steel [C3FS03-04]. The responsible engineer must establish the project welding basis and edition; neither listing automatically approves this attraction. HSS joint forces are nonuniform, so a continuous-looking weld is not proof of an adequate joint [C3FS05].

The fabrication traveler follows this sequence:

1. Resolve jurisdiction/classification, design standards, permit route and project QA responsibilities. If the county ARPM chapter applies, coordinate its engineering, QA, inspection and commissioning requirements; these records are not an approved program [C3FS09].

2. Release material/toughness, complete joints, floor/seat hardpoints, draft loads, fatigue, OEM interfaces and handling cases. Approve WPS/PQR or applicable prequalification, personnel and inspection plan.

3. Receive, identify and quarantine stock as needed; preserve heat/certificate mapping through cutting. Record actual wall, condition and seam orientation.

4. Establish approved fixture contacts and A/B/C datums. A is steel bearing z31.003937; B is y0 from mains y±34; C is front face x−47, 47 in before cabin x0.

5. Dry assemble mains/bolsters, then transverse/perimeter/seat rails. Confirm weld and inspection access before closing any compartment. Survey before and after tack welding.

6. Weld in the approved balanced sequence, recording heat control and distortion checkpoints. Stop for departures; do not improvise corrective heating or repair welds.

7. Complete specified inspection before coatings or concealment. Resolve nonconformances and resurvey the free frame after fixture release.

8. Trial-fit approved interfaces and service-removal paths. Coordinate coatings, dissimilar-material isolation, drainage and adhesives. If galvanizing is chosen, design rectangular-HSS venting with the galvanizer and structural engineer [C3FS07].

9. Install qualified floor/body/systems, weigh components and the complete coach, reconcile per-wheel reactions/CG, and issue as-built evidence before separately authorized commissioning.

Survey points P1-P4 at x45/261 and y±34 are measurement locations, not fixture or lifting points. Their 216×68 in rectangle has nominal diagonals 226.450878 in. Record three-dimensional coordinates: diagonal equality alone does not prove flatness or straightness. Twist diagnostic is (zP1−zP2)−(zP3−zP4). All allowable deviations, clamp forces and acceptance signatures remain blank; building-fabrication tolerances are not automatically vehicle tolerances [C3FS08].

Handling cases include the 1,527.288 kg bare frame, 7,040 kg empty allocation, and a 5,240 kg body-only allocation after subtracting allocated bogies. Actual weights, lift/jack points, unequal support reactions, bogie retention, temporary bracing, rigging and transport-restraint accelerations need specialist design. No lifting lug, jack contact, sling angle or transport tie-down is released.

Farm crews can maintain traceability, stage stock, assist calibrated measurements, perform approved nonstructural fitting/coating preparation and assemble released noncritical items under instruction. Vehicle structural design, frame welding, critical interfaces, NDE, brake/electrical installation, lifting plans and acceptance require the designated specialists. A competent farm welder may participate only within the same project qualifications, procedures and inspection controls.

Five editable CSV records accompany this module: fabrication traveler, 110-member material traceability, joint-family release, datum survey and mass/CG measurement. Empty acceptance fields mean unresolved work. The joint-family form is a starting register, not a complete weld map; the engineer/fabricator must assign every actual joint before fabrication.

04 | Running gear, connections and locomotive demands

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The next useful release is a **matched locomotive, coupler, bogie and brake interface package**, followed by a controlled first-coach installation. The calculations below expose the necessary space and force returns. They do not establish equipment ratings. F1 retains the E1 frame at a **32-inch floor as an integration study**; 36-inch gauge, 65,100-kg upper train mass, 50,600-kg trailing mass and the owner's **0% longitudinal grade** remain the calculation basis. Nominal mass allocations are unchanged.

Coupling geometry that can be drawn now

Each study coach has a 400 ×110-inch frame, bogie centres 216 inches apart, and a pivot 14 inches inside each end. A 30-inch straight platform gap therefore gives a 58-inch pivot span. These are sensitivity dimensions, not an approved coupler configuration. Actual coupling height, stroke, width, joint details and mounting dimensions remain blank. No passenger gangway across the gap is assumed.

For two identical coaches on one constant-radius circle, let the half-frame length b=200 inches, half-width w=55, pivot arm a=186, span d=58. With radius R and proxy bogie wheelbase 1165 mm, the body-centre radius is $Rc=\sqrt{[R^2-(1165/2)^2-(216\times25.4/2)^2]}$, in millimetres before conversion to inches. The half-angle between bodies is $t=\operatorname{atan}(a/Rc)+\operatorname{asin}[d/(2\sqrt{(Rc^2+a^2)})]$. The facing inner/outer corner separations are $d-28\cos(t)\mp110\sin(t)$. A separate convex-polygon distance calculation reproduces the inner gaps.

Constant-radius case	Coupler angle to each coach	Relative body heading	Inside corner gap	Outside corner gap
Straight	0°	0°	30.00 in	30.00 in
30 m	10.384°	20.768°	10.63 in	50.29 in
50 m	6.248°	12.496°	18.20 in	42.14 in
75 m	4.169°	8.338°	22.08 in	38.07 in

The 30-m case is particularly demanding for guards, cables and end gates. The link centreline crosses the end-frame plane 2.565 inches sideways from its straight position; that is **centreline travel only**, before coupler width, draft stroke and clearance. Reducing the effective pivot span from 58 to 54 inches gives only 6.81 inches at the inside corner on30 m. That is an analyst pose comparison, not an assumed 4-inch damper stroke. Reverse curves, tangent transitions, dissimilar locomotive geometry, worn slack, track twist and suspension movement still require a full multibody envelope. These positive rectangular gaps are not an accepted minimum radius or passenger clearance.

The saved results contain exact plan polygons for both full frames, facing platforms and the boarding team's deployed side-stair study. Coordinates use a common origin halfway between coupling pivots. Deployed stairs are a stationary boarding interface; a safe stowed moving envelope has not been established.

Rail coupling and installation interfaces

Voith Type 430/530 is a compact folding rail-coupler family with published tension/compression strength **up to 300 kN**; Voith also offers semi-permanent connections with energy absorption and services [F-M01]. This is family strength, not this project's working or fatigue rating. Dellner's people-mover semi-permanent coupler identifies a double-acting centre damper, pivot anchor and cable support, without a numerical installation envelope or load rating [F-M02]. Both are real rail-only engineering routes; neither has demonstrated an under-platform fit here.

The drawing's exploded view is functional: frame diaphragm/anchor → articulating mount → draft/damping function → retained separable joint → opposite articulating mount/anchor, with a separate protected service cradle and passenger barrier. Functions may be combined in one OEM assembly. Block spacing is illustrative; it does not prescribe part count, pins, welds or lengths. Anchors must transfer reversing longitudinal force, transverse components and offset moments into both main girders through an engineered load path. Platform floor and guard members cannot simply become draft anchors.

Installation sequence: accept the OEM interface drawing and load cases; fabricate/inspect the released frame attachment; verify datum/height with unloaded and loaded vehicles; install the complete retained coupling using OEM assembly procedures; route and support services through the **entire** approved yaw/pitch/roll/stroke envelope; install guards without trapping cables or preventing inspection; perform documented retention, release, separation-detection and brake tests. Coupler removal direction, tool access, lifting support, fastener torque, lubrication and wear gauges must come from actual hardware instructions. Before uncoupling, secure both vehicles independently and isolate stored energy under an approved maintenance procedure.

Passenger gates must exclude the changing inter-car gap and pinch zone. Jumper continuity does not prove the mechanical joint is intact: automatic breakaway braking must detect the relevant mechanical separation even if cables remain connected. The electrical/brake module controls the monitored emergency function. Do not infer hidden coupling engineering from Western Train photographs.

First-train force, power and braking

The portable calculator uses F=M(a+gc)+Fextra at zero grade. Starting-resistance coefficient 0.010 and running 0.005 are retained analyst screens, not measurements. Additional 0/2/5-kN resistance rows represent unknown curve/wind resistance; **no radius-specific resistance law is invented**. Applying starting coefficient at speed is a bounding resistance sensitivity, not a claim that starting drag persists in service.

Starting-coefficient sensitivity	Wheel force	Wheel power at 7.5 mph	Engine screen*	Driven mass at assumed μ 0.15
a=0.15 m/s²	16.15 kN	54.14 kW	82.19 kW	10.98 t
a=0.30 m/s²	25.91 kN	86.88 kW	125.85 kW	17.62 t
a=0.50 m/s²	38.93 kN	130.54 kW	184.05 kW	26.47 t

*75% drivetrain efficiency plus 10 kW propulsion auxiliaries; separate hotel power excluded. These accelerations are sensitivity cases, not operating requirements. At μ 0.15, even the entire 14.5-t locomotive/hotel allowance provides only 21.33 kN adhesion. Its theoretical starting acceleration ceiling is 0.2296 m/s² before extra drag; the actual powered fraction will be smaller if hotel/tender mass is unpowered. A 5-kN extra resistance allowance adds 22.35 kW to the7.5-mph engine screen. More horsepower alone cannot cure insufficient adhesive weight.

Cummins B4.5 173 publishes 129 kW at 2000 rpm as **intermittent**, with 780 Nm peak torque at 1500 rpm [F-M07]. This identifies a real candidate calibration, not selected continuous capability. At2000 rpm its power corresponds to 615.93 Nm, so the1500-rpm peak torque must not be substituted. Illustrative 500/600/800-mm locomotive wheels require total equivalent reductions 15.62/18.74/24.99 at 7.5 mph and 2000 rpm; no transmission or axle is selected. Require duty definition, site derating, aftertreatment/application acceptance, complete torque maps and cooling rejection before sizing the

driveline.

SCHÖMA CHL-60G provides a documented 12-16-t hydrostatic comparison, approximately 55-100 kW, gauge≥600 mm and approximately 25 km/h; its standard catalog train brakes are pneumatic [F-M06]. A 16-t configuration already exceeds the locomotive/hotel allocation. Severn Lamb JUPITER identifies 914 mm gauge,50-m minimum radius and 16 km/h [F-M05]; the30-m study does not meet that published radius. Neither source provides a usable project drawbar curve. Irwin's manufacturer tractive-effort chart lists 5,000/7,500/10,000 lbf for its 10/15/20-ton columns, but no exact model or speed-dependent conditions [F-M08]. Those figures cannot be converted into an accepted train capacity.

Prefer a complete engineered diesel locomotive offer. Diesel-mechanical requires a controllable launch/reversing transmission; hydrostatic offers low-speed modulation with a consequential oil-cooling duty; diesel-electric separates engine speed from wheel speed but adds generator, motor and control packaging. Compare guaranteed wheel/drawbar curves, powered axle weights, forward/reverse behavior and heat balance - not engine output alone. Hydraulic propulsion does not satisfy the separate hydraulic-brake preference. Maintain that preference pending the coordinated brake safety case; catalog pneumatic brakes are an alternative requiring evaluation.

At the first connection, launch with a=0.15/0.30/0.50 m/s² and c=0.010 produces 12.55/20.14/30.26 kN longitudinal tension when pulling and compression when pushing in the ideal model. For the identical-coach 30-m angle, the 0.50 case becomes 30.77 kN along the link and 5.55 kN transverse at an anchor. **The actual locomotive-to-coach angle is unknown**; these components are a geometry sensitivity, not its solved load case. Full train dynamics must add slack, brake delays, loads during faults and longitudinal stability under compression.

Brake demands at 0.30/0.50/0.80 m/s² are 19.53/32.55/52.08 kN. At7.5 mph, three seconds before constant deceleration gives 28.79/21.30/17.08 m stopping distance; at 5 mph,15.03/11.70/9.83 m. Translational energy is 365.90/162.62kJ at 7.5/5 mph. No rolling-resistance credit, rotating inertia, application ramp or adverse adhesion is included. Equal mass-proportional simultaneous brakes give zero ideal coupling load on straight track; real imbalance does not. Locomotive-only μ 0.15 braking yields just 0.3276 m/s² and 16.58 kN first-joint compression in the idealized case, which is not an acceptable substitute for coordinated car brakes.

The first joint's two-group reduced mass is 11,270 kg. Relative closing speeds 0.10/0.25/0.50 m/s contain 56/352/1,409J before any impact absorption. These are sensitivities, not permitted coupling speeds. Actual force-stroke data and dynamic analysis are needed to convert energy into peak loads; guessed pin and weld sizes would hide this unresolved requirement.

Bogie and track work that can proceed

F1's nominal bolster underside is 482.7 mm above rail. OEM loaded/unloaded support height, attachment stack, sidebearer preload, new/worn wheels, stops and travel must jointly satisfy that datum. Railteco RTKZ1 is a **rejected supplied assembly**: 762-mm gauge and 3.75-t axle rating do not meet this evaluation geometry and retained 8.295-t front-bogie gross reaction [F-M03-04]. Its 596-mm wheel / 1165-mm wheelbase remain identified comparison dimensions only.

Retained P3 wheel-cylinder sampled union boxes extend x 7.279-82.721 and 223.279-298.721 inches, y±23.535, z−0.508-27.981. Negative z belongs to an inflated mathematical proxy. These are not complete-bogie keepouts: brakes, frame, axleboxes and services are missing. The proposed central x 93-213 service reservation has 10.279 inches longitudinal separation from those proxy boxes only. It is not established usable space. Freeze OEM wheel/rail profile, back-to-back, support/sidebearer and all motion/removal volumes before chassis installation release.

The farm crew can now organize controlled survey observations, existing drainage/utility mapping, receiving records and ground-level mockups. Qualified survey/civil/geotechnical/track staff must establish the elevation datum, soil/compaction evidence, drainage outlets, ballast/subgrade, rail/tie/fastener design, curve/turnout geometry, crosslevel/twist tolerances and crossing/access provisions. After issued work packages, farm earthwork, drainage, ballast and tie/rail placement can proceed through inspection hold points. A 0% longitudinal rail target still needs drainage and measured settlement/crosslevel acceptance. Do not finalize alignment or assume reverse-curve transitions before vehicle interfaces and the deferred North Pole location are resolved.

The saved input/result/register files preserve every sensitivity and missing return. Public sources were checked 18 October 2026 UTC; no supplier contact, availability confirmation, quote or project approval is claimed.

05 | Boarding, platforms and evacuation interfaces

PF-TRAIN-S3F-R0 · 18 October 2026 UTC · **F1 = E1 at a 32-inch floor, study only. No final floor selection, fabrication release or elevated occupied test.**

Develop normal boarding around near-level station platforms and engineered retained bridges, while preserving the cabin. Continue the onboard stair requirement as a side-descending arrangement whose stowage must be engineered. A full fixed flight projecting from the car is unsuitable for the retained moving-envelope study. External station lifts can preserve interior space, but the investigated products expose important capacity and interface limitations. Neither a bridge nor an onboard lift alone resolves every wayside evacuation condition.

The cabin remains 306 × 104 inches clear. The 44-inch end platforms, frame x = −47 to 353, bogie centres x = 45/261 and 30-inch intercar gap are unchanged. The gap is an articulation sensitivity, not a selected connection. Coordinates below use inches, with z = rail top. Ground mockups remap every walking surface onto the existing floor at z = 0.

Side stairs fit the platform plan, but cannot remain deployed in motion

Reserve a **36-inch clear stair flight** at front x = −43 to −7 or rear x = 313 to 349. It descends outward from y = +55; mirror it for the opposite side. This leaves four inches gross at each longitudinal end of the 44-inch platform. That is space for investigation, not a proven allowance for posts, hinges or guards. Stair side hardware belongs outside the clear width and may consume those margins.

Use the conditional building-stair benchmark of a maximum seven-inch riser and eleven-inch going to compare envelopes. Final onboard requirements depend on the accepted vehicle and egress standards. [C3FB12]

Floor above rail	Actual landing above rail	Total rise	Risers × uniform rise	Treads / outboard run
32	0	32	5 × 6.4	4 / 44
32	−4	36	6 × 6.0	5 / 55
32	+4	28	4 × 7.0	3 / 33
36	0	36	6 × 6.0	5 / 55

For the F1 front/right case, the four walking surfaces occupy x = −43 to −7, with y/z pairs **55-66/25.6; 66-77/19.2; 77-88/12.8; 88-99/6.4**. The lower landing trial is x = −43 to −7, y = 99-135, z = 0. Its 36-inch depth is a maneuvering study, not an adopted landing minimum. Plan, side and exploded concept coordinates are in boarding_f.json; tread thickness, strings/supports, hinges, pins, welds, guard heights and anchors remain null.

A fixed five-riser flight matched to a zero-elevation landing has a 6.4-inch bottom rise. Changing the landing to +4 or −4 changes that bottom rise to 2.4 or 10.4 inches while the other risers remain 6.4. Passenger loading and suspension motion add further variation. Therefore coordinate an actual landing elevation and the stair's supported/unsupported design; do not assume that level track creates a uniform stair at arbitrary ground locations.

The lowest F1 tread is 6.4 inches above rail before its unknown thickness. It lies approximately 12.60 inches below the retained F075/12-inch-frame underside. Its underside is consequently lower still. Actual crossing, tie, ballast, rail, turnout and suspension clearances must be checked with the complete stowed equipment.

No under-platform cassette or hinge location is selected.

The deployed flight reaches |y| = 99 inches at a 32-inch floor, or 110 inches at 36 inches, versus the body's 55-inch half-width. A rigid-body curve screen at 30 m radius gives outside excursions from track centreline of **1.681 m for the body, 2.769 m with the 32-inch-floor deployed flight and 3.045 m with the 36-inch-floor flight**. These exclude dynamic movement, guards, tolerances and connection constraints. They demonstrate why the deployed footprint cannot become the running configuration.

End-facing stairs are rejected: a 44/55-inch run exceeds the retained 30-inch gap by 14/25 inches even on straight track and intrudes into connection space. The mechanical module's equal-radius pair study reduces the inner platform-corner gap to about 10.63 inches at 30 m radius. The adjacent inside deployed stair rectangles are only 2.639 inches apart at that radius, versus 38 inches on straight track. Guards, mechanisms and passengers are excluded, so even stationary boarding needs the actual station alignment and complete deployment envelopes. Side stairs avoid routing passengers into the centre connection, but actual stair/guard/coupling sweeps still require coordinated verification. End edges need independent passenger barriers; no passenger gangway is selected.

Rail equipment is available, but the full flight needs a configured proposal

IFE STR and Wabtec Access sliding devices are real rail products, with published installation heights of 56 and 57 mm respectively. Their example masses are 65 kg, but these examples do not define this project's stair mass or rating. Compared with the retained 25.3 mm floor stack, a flush installation would extend another 30.7/31.7 mm into the structural zone before mounting details. They are gap/single-step devices, not a demonstrated 32-inch ground staircase. [C3FB01, C3FB03]

BODE BCBS offers configurable multistage motion and monitoring; it is a useful supplier lead for the full arrangement. No retrieved document establishes a 36-inch-clear, five-riser package fitting these end platforms. Obtain its actual stowed envelope, mechanical retention, load reactions, passenger detection, emergency operation and service access. A vehicle designer must accept the combined stair/platform/frame design. [C3FB02]

A detached rated station stair is another way to avoid moving-envelope interference. It needs secured docking, correct landing height, guards and dispatch removal checks. It does not satisfy the onboard-stair objective or provide independent evacuation between stations. Do not replace either function silently.

Near-level station interface and motion tolerance

At the accessible side door, retain x = 9-51 and the 42-inch clear opening. Keep the prior **36 × 36-inch operating reservation** at x = 12-48, y = 55-91, with a 60 × 60-inch station-turn reservation at x = 0-60, y = 91-151. Mirror for the other side. A nominal station-edge study at y = 58 leaves a three-inch body gap; nominal station level equals the 32-inch floor. Neither is a final station dimension.

The operating reservation is not a fabricated plate. Actual bridge length must reconcile the real gap, support overlaps and usable sloped run. Bearings, retention, plate thickness, edge protection, loaded deflection and stowage need design. A friction-only loose plate is not an accepted solution. The normal bridge must also tolerate the specified stopping-position and vehicle-motion range.

An explicit sensitivity combines one inch of centre heave with side-edge movement at a 55-inch half-width:

Assumed roll	Combined vertical mismatch	Horizontal projection at 1:12
0°	1.000 in	12.00 in
1°	1.960 in	23.52 in
2°	2.919 in	35.03 in

These are analyst cases, not actual suspension limits. Track/platform tolerances and settlement remain additional. Where amusement §1002 applies, direct boarding gaps are assessed under normal passenger loading; the bridge exception invokes applicable vehicle-device provisions. A nominally level drawing is insufficient. [C3FB09-10]

A ground ramp at 1:12 requires 32 feet of horizontal projection for F1, or 36 feet for the higher comparison, before landings. Where facility §405 applies, the 30-inch maximum rise per run means either case requires at least two runs. This is a different installation from the small normal station bridge. The applicable ramp branch and final layout remain for review. [C3FB13]

Accessible lift comparisons reveal real limitations

Use actual combined occupant, device, baggage and any attendant load. Equal division of the retained 300 kg two-device allowance gives a **150 kg device sensitivity**, not an admission limit. Adding a 145 kg adult gives 295 kg; adding a 100 kg attendant gives 395 kg.

Reference	Verified capability	Project consequence
Adaptive Engineering Mobilift TX	Manual station rail lift; 600 lb rating; 60 in lift; 34 × 56 in platform	600 lb is 272.2 kg, below the 295 kg sensitivity. Published 29 in bridge width and conflicting unit masses require clarification.
Guldmann Stepless LP11	300 kg rating; 180-955 mm lift; 149 kg unit mass	Only 5 kg remains above 295 kg before baggage; no 100 kg attendant allowance. Height margin is 142.2 mm at F1 and 40.6 mm at 36 in.
BraunAbility NCL1000IB3754HB-2	1000 lb rating; 48 in travel; 37 × 54 in platform; 385 lb unit mass	Rating is 453.6 kg, but retained 42 in doorway is 3.85 in narrower than its 45.85 in minimum installation opening. Rail application is unaccepted.

These are configured references, not purchases or approval of the passengers this train can accommodate. LP11 requires stable even support, trained operation and a stationary train. Mobilift's literature prohibits moving its lift while occupied; its approximate transport dimensions are not a proven all-terrain rescue envelope. [C3FB05-08]

The BraunAbility threshold reservation remains x = 4.52-55.48, y = 34-52. It overlaps the retained turning circle by two inches. Moving the circle centre from (30,6) to (30,3) produces only one inch of mathematical separation; this does not clear hardware or a real device. The referenced guide also reduces vertical passage by 1.375 inches relative to the door opening. Keep exact stowed/deployed CAD, mounting loads and manual recovery under review. Its 12 VDC/70 A maximum is a candidate electrical datum, not selected onboard demand. [C3FB04]

Preserve the existing 78-inch door study while evaluating an **80-inch clear-opening alternative**. The ADA closer/stop exception does not establish a general 78-inch structural doorway. Facility lifts and vehicle lifts have different requirements; the applicable category must be resolved before adopting any product compliance claim. [C3FB11, C3FB13, C3FB15]

Dispatch logic and rescue must work with the physical equipment

The JSON specifies functional states rather than a certified control circuit. Deployment requires independent standstill, approved brake holding, traction inhibition, correct-side authority and a checked landing. Release the passenger gate only after the appropriate stair lock/support, retained bridge, or lift landing/barrier conditions are established. Door commands alone are not physical position proof.

Travel permission requires every passenger door/gate closed and latched, every aid stowed with retention proven, and no boarding lockout or unresolved fault. Loss of power must remove travel permission without automatically retracting an occupied aid or releasing a holding brake. Controlled manual emergency egress and recovery must remain available under the accepted evacuation plan. The brake/control integrator specifies fault response, monitoring, diagnostics and override discipline.

Unknown North Pole location does not prevent interface engineering, but it prevents a complete rescue-coverage claim. Compare controlled recovery to a safe station; adequately rated mobile lifts at reachable stable pads; a designed ramp with a continuous escape route; and trained assisted transfer using suitable equipment. A fire or damaged vehicle may prevent recovery by rail. Both sides, blocked exits, lighting, communications, equipment delivery and device recovery need route-specific planning. Carrying

passengers by hand is not a default solution.

The current county basis is Ordinance 5378, effective 3 June 2026. Refer to the coordinated jurisdiction update: classification and referenced editions remain held. [C3FB14] Before occupied boarding trials, qualify the actual installed equipment under the Phase 3E E14 sequence, then conduct supervised participant work. Farm crew can now tape these footprints and measure real devices at ground level; provisional stairs, bridges, lifts and elevated decks are not authorized for people.

06 | Brakes, controls, HVAC and electrical installation

PF-TRAIN-S3F-R0 · 18 October 2026 UTC · **F1 installation study: 32 in floor. No fabrication, energization, pressure settings or equipment approval.**

Develop the first regular coach around two local hydraulic brake zones, a separate brake reserve, independently backed emergency lighting and an externally serviced HVAC system. Keep the cabin free of equipment cabinets. The generator and its exhaust remain on the locomotive. The saved package advances the installation interfaces: 17 space reservations, 12 circuit/port returns, 17 proposed control transitions and separate functional schematic layers. None is an OEM-approved fit.

Hydraulic service, emergency and parking architecture

Retain hydraulic service braking. The preferred inquiry family uses spring-applied, hydraulically released actuators with controlled release pressure for service modulation, provided the matched brake supplier accepts repeated modulation, empty/full balance and the complete fault envelope. Hanning & Kahl documents spring application and configured hydraulic units; those product families do not establish this installation's force, flow, pressure or current. [C3FU04-05]

Each bogie receives a local pressure-generation/regulation module and an independently monitored emergency application function. The ordinary service ECU commands modulation; an emergency/separation function also removes locomotive traction permission and requires both bogies to apply. Local mechanical springs provide application energy without hotel AC or a functioning battery. **The battery supports controls and release functions; it is not the stored emergency braking energy.**

The schematic represents required functions, not a valve circuit. The supplier must prove how release pressure is removed after power loss, a stuck valve, blocked return, hose failure and trapped auxiliary pressure. Stopping the pump alone does not prove application. A second hydraulic release circuit can hold a spring brake released independently of the service circuit; recovery isolation and restoration therefore require positive indication and controlled procedures. [C3FU06]

A shared actuator, disc, mount or wheel remains a common-cause limitation. Calling the spring store independent does not create a second independent braking system. If service-modulation or fault evidence is inadequate, compare active hydraulic service with a separately engineered spring emergency/parking actuator, or a complete automatic-air package. Knorr-Bremse documents both hydraulic construction families, but no mixed-supplier assembly is qualified here. [C3FU08]

Retain the 13,200 kg regular-coach upper analysis case and the earlier 0.50/0.80 m/s² service/emergency sensitivities; these require 6.60/10.56 kN before rotational, adhesion and dynamic effects. Actual axle loads, wheel radii, force maps and stopping criteria still control selection. Zero design grade does not eliminate parking disturbance, achieved-level tolerance or recovery restraint requirements.

Installation zones and maintainable routes

Coordinates are inches: x from the front inside wall, y from the centreline, z above rail. F1's finished floor is 32; the inherited ceiling is 118.614 and the flat roof proxy is 121.614. Crown, curb and final roof structure remain unresolved.

Reservation	x	y	z	Installation consequence
Front / rear hydraulic modules	96-132 / 177-213	−29 to −3	18-25	Only 7 in height; no actual module fit established
Brake reserve / DC panel	138-168	−29 to −3	18-25	Dedicated brake enclosure; battery type and ventilation held
AC distribution / conversion	96-132	3-29	18-25	Transformer and protective-device fit unknown
Light reserve / hotel controls	144-180	3-29	18-25	Separate from brake reserve
Generic matched roof HVAC	88-218	−35 to 35	121.614-145.614	130 × 70 × 24 space allocation, not an OEM envelope
Two external supply ducts	21-285	−53 to −35 / 35-53	121.614-136.614	Each 18 × 15 gross; above occupied space

The structural module confirms only nominal frame-box nonintersection. Complete bogie motion, brake gear, debris protection and support loads remain unverified. Side withdrawal at the installed box elevation crosses the main girders, whose underside is 19.003937 in. Lower the entire box beneath that underside plus an engineer-set clearance before lateral removal, or provide an engineered pit/lifting arrangement. Separate maintenance-only withdrawal volumes are supplied; they are not working-clearance approvals or lifting instructions.

The central-span routing reservations are hydraulic y = −31 to −27, controls y = −2 to 2 and AC y = 27 to 31, each x = 75-231 and z = 25-26.5. They deliberately stop before unresolved bolster/bogie crossovers. Physical partitions, drainage, fastening and final electrical separation are held; the drawn gap is not a code clearance.

Use protected fixed tubing on the body with OEM-designed flexible loops at moving interfaces. The line schedule must record exact hose/fittings, fluid, temperature/pressure transients, free length, bend radius, orientation, clamps and full-motion sweep. Parker's guide addresses these factors and inspection access, but supplies no universal train hose selection. [C3FU07] Keep front and rear zones identifiable; no intercar hydraulic service hose is proposed. Intercar AC, service data and emergency functions require separate reviewed routes and retention/separation behavior.

Control states to resolve before wiring

The full matrix retains actual I/O, timeouts and stop profiles as blanks. The key proposed transitions are:

Event	Required response	Return to service
Power-up or reboot	Applied/parked state; traction inhibited; verify actual brake state	Deliberate reset and release request after self-check
Boarding	Standstill and parking hold; accepted stair support/lock, retained bridge or lift landing/barrier state before gate release	All aids mechanically stowed/retained or detached and clear; doors/gates latched; explicit dispatch
Boarding-aid power loss	Maintain holding brake; no automatic retraction of an occupied aid; retain accepted manual recovery/egress	Restore and reprove support, locks, barriers and readiness
Door/gate opens in motion	Remove traction and request emergency response through independent function	Secure, inspect, restore and reset; stop profile needs hazard/OEM review
E-stop, separation or invalid service data	Latched emergency request and traction inhibit on affected consist	Identify cause and prove restoration; data recovery alone cannot release

Event	Required response	Return to service
Hotel AC loss	Hotel loads stop; independent lights operate; controlled stop and secure	Staged restart; no brake release on source return
Brake DC loss	Application must follow through accepted stored-energy architecture	Restore supply, close fault and deliberately reset
Reversing	Require accepted zero-speed evidence and hold; requalify active cab/direction	Release only after all readiness conditions
Shore connected or recovery release active	No normal passenger movement permission	Disconnect/restore, inspect and reprove functions

Gate closure, E-stop reset and source restoration are distinct from brake release. Door monitoring must preserve manual emergency escape. Electrical jumper continuity alone cannot establish that a mechanical coupling remains intact. A bypassed bogie or auxiliary-released brake does not authorize continued passenger service.

Separate winter and hot-occupied duties

The updated calculation uses the retained geometry and window schedule, 32 passengers plus one person allowance in each regular coach, and 13 passengers plus one in the accessible coach: 113 thermal occupants for the consist, not a staffing prescription. Outside air remains an **assumed** 7.5 L/s per person plus 0.03 m³/s average door leakage. Whole-body opaque U = 1.5 W/m²K is a planning target, with 0.7/2.5 sensitivities; it is not demonstrated by the prior insulation build-up. Glass U = 2.7 and SHGC = 0.5 are also unselected assumptions.

Thermal case at working U	Regular coach	Accessible coach
Christmas evening: 30°F outside / 68°F inside, occupied net heat	7.37 kW	5.11 kW
Same conditions, loss before internal gains	10.04 kW	6.36 kW
Colder 20°F evening, occupied net heat	10.01 kW	6.78 kW
Hot 110°F / 75°F, dry evening without solar	15.42 kW	8.90 kW
Hot dry day with stated solar	18.61 kW	11.52 kW
Hot day plus humidity-ratio stress	24.36 kW	14.32 kW

Cooling includes a 15% analytical allowance. The humidity stress adds Δw = 0.006 kg/kg; it is not a claim about typical Moapa Valley weather. Solar is separately visible. Heating subtracts occupant sensible and 200 W internal-equipment heat per coach, but includes no warm-up capacity or infiltration gust allowance. That 200 W is a separate thermal allocation, not the 1,080 W electrical auxiliary total: fan/lighting/control heat entering the cabin and underfloor charger losses still need reconciliation. Actual climate design conditions, envelope bridges, ventilation requirements, door cycles and thermal mass must be resolved before equipment selection. At U = 2.5, regular-coach hot/humid duty rises to 26.58 kW and 30°F occupied heat to 9.02 kW.

At the working hot sensible duty and a 12 K supply-air difference, the regular coach requires approximately 1.167 m³/s in this screen. Two external 18 × 15 in gross ducts give 3.35 m/s. A 4 m/s analyst velocity objective requires at least 0.292 m² **net** area, compared with 0.348 m² gross allocation. Liners, insulation, balancing, bends and acoustics consume that margin. The ducts sit above the roof, preserving cabin headroom while materially changing the exterior envelope. Actual condenser inlet/discharge clearance must be checked against the adjacent ducts. Return-air, fresh-air and branch layouts remain held; these boxes are not a complete duct design.

HVAC and electrical architecture tradeoff

Keep two electrical alternatives open. **B is preferred for a configuration inquiry:** dedicated 480 V three-phase rail HVAC with a designed 120/240 V auxiliary secondary. A uses 120/240 V split phase for all hotel loads. Northwest Rail Electric's NW-4 normally uses 480 V three phase; other voltages require a configured offer. Its family maxima do not establish a suitable coach unit. The NW-2 undercar package has no demonstrated fit in the 7 in reservation. [C3FU02-03] Retained Konvekta 400 V references cannot be connected directly to either study bus by assumption.

The current Dometic manual supplies a documented modular comparison. Both FJX3573M and FJX5557M nominally provide 15,000 Btu/h at 115 V, 60 Hz, single phase. The former lists compressor/fan rated currents 13.6/2.6 A and locked currents 71/6.9 A; the latter 12.8/3.6 A and 57/8.2 A. Their intended RV application does not establish rail acceptance or 110°F delivered capacity. [C3FU01]

Six modules provide 26.38 kW nominal cooling per regular coach. At an **assumed** 90% retained capacity they fall 0.62 kW below the working hot/humid screen. Six FreshJet 3 units total 11.18 kVA at nameplate voltage; one compressor starting while the others run gives a 17.78 kVA scalar screen and 106 A on the starting leg. Locked-rotor current is not a measured start waveform. Auxiliary loads are additional.

Roof-space consequences are substantial. Six FreshJet 5 bodies total 242.34 in length; adding one analyst 18 in service/air interval per unit creates a 350.34 in serial study, exceeding the 306 in cabin length by 44.34 in. This is a rejected serial arrangement, not a manufacturer's spacing requirement. FreshJet 3's corresponding 285.6 in study leaves little room for end access, roof framing and other services. Alternative bank layouts, mutual airflow, noise, roof penetrations and controls require fresh review. The mechanical air-distribution box also projects into the cabin; no retained-headroom modular installation is shown. The external-duct headroom statement applies only to the rail-package allocation. **Do not purchase six units or cut roof openings from this comparison.**

At assumed COP 2.5, the working regular hot/humid demand gives 9.74 kW HVAC input plus 1.08 kW auxiliary allocation. Architecture A's balanced 240 V screen is 50.10 A at assumed PF 0.90. B is 14.54 A balanced-equivalent at 480 V after an assumed 95% efficient auxiliary conversion; actual single-phase secondary loading requires phase-by-phase analysis. These are calculated allowances, not nameplate currents. The calculator retains COP and starting sensitivities separately.

For all three regular coaches plus the accessible coach, the hot/humid solar case at COP 2.5 totals **39.28 kW on A or 39.50 kW on B**, including auxiliary allowances. Across COP 2-3 the corresponding A screen spans 48.02-33.45 kW. At 30°F, ideal resistance heating gives 31.53/31.76 kW occupied A/B demand; loss before internal gains gives 40.81/41.03 kW. **Neither is complete warm-up demand.** Effective body thermal capacitance, initial temperature, target warm-up time and initial ventilation remain blank; additional warm-up input and the generator selection remain unresolved.

The conceptual one-line includes interlocked shore/generator selection, protected trunk, local isolation, HVAC branch and auxiliary panel. Brake and lighting chargers have separate protected branches and reserves. Phase/neutral/PE arrangements, earth bonding, fault current, disconnect contact order, flexible cable ratings and selective protection remain electrical-engineer decisions. Frames and running rails are not assumed normal current-return paths. No conductor, fuse, breaker or connector rating is released.

DC allowances and first-coach handoff

The nominal 24 V brake budget allocates 6 A continuous nonpump demand, two 20 A pump loads at 10% duty, 10 A average and 46 A with both pumps running. Assumed starting multipliers give 46/86/126 A sensitivities. These are deliberately labeled allocations; the HCU configuration must replace them with documented currents and voltage limits. One hour at the assumed average with 60% usable capacity gives 16.67 Ah; chemistry, cold performance, aging, discharge capability and peak voltage sag remain unresolved. Emergency application must not depend on that duration.

A separate 20 W emergency-light allowance gives 1.39 Ah at 24 V for one hour with the same assumed usable fraction. This establishes no required illumination or evacuation duration. The accessible coach's optional lift is a separate conditional supply interface, never a load on the brake reserve; no lift demand is added as selected equipment.

Five new installation work packages assign actual fit/access closure, matched brake/control design, source/circuit engineering, thermal/duct completion and installed first-coach verification. Blank forms capture nameplates, ports, state behavior, installation envelopes and thermal conditions. Continue the retained Phase 3E tests after released installation, including installed boarding qualification before participant exercises. Farm crew can label and prepare protected routes under released details; specialist integrators own hydraulic assembly, energized work, controls and acceptance interpretation.

07 | Files, coordination and limits

Format / record	Use
PDF / SVG drawings	Review installation relationships, interface dimensions, schematics and work sequence. Omitted safety-critical dimensions remain explicitly held.
DXF geometry	Use declared actual-inch geometry or separately identified paper-inch sheet exports. Do not scale a sheet to invent a missing connection or manufacturing dimension.
STEP / browser mesh	Nominal geometry and colored installation reservations. Reserved space is not a selected component; shown wheel cylinders are not a full bogie or dynamic envelope.
XLSX / JSON / scripts	Traceable calculations, allocated budgets and blank actual records. Native engineering acceptance and physical measurements are separate from arithmetic/export checks.
Fabrication and interface records	Capture issued drawing revisions, material evidence, weld/inspection requirements, measurements, deviations and responsible approvals before work advances.

Physical work for the farm crew remains conditional on the released scope: ground-level mockups and information gathering can proceed from the defined instructions; vehicle structure, pressure systems, safety controls, electrical protection, lifting, elevated boarding structures and controlled vehicle testing need specialist design and supervision. Track construction follows surveyed and released track work packages; the 0% target alone does not define a buildable track plan.

A final issue review checks calculation/source consistency, drawing coordination, model exports and readable document rendering. The release record states the scope of any site/model checks. No physical prototype test or engineering approval is implied by publication.

08 | Primary sources and retained calculations

Sources were reviewed on 18 October 2026 UTC unless another date is shown. Access dates and limits accompany the editable source register. Public manufacturer data does not establish stock, price, delivery, rail application approval or compatibility of the offered configuration. Referenced legal and code provisions are conditional until the responsible authority establishes jurisdiction and applicable requirements.

FJ01 | Clark County signed Ordinance 5378 - Chapter 22.16

Signed ordinance; effective June 3, 2026. Scope, classification, engineering, permit, inspection and commissioning provisions read; Appendix A checked.

Limit: Project classification, exemptions and applicable standard editions require responsible authority confirmation; no authority contact or approval claimed.

https://www.clarkcountynv.gov/adobe/assets/urn:aaid:aem:c5347a68-d8cb-444a-9cf2-e9bf7b12080f/original/as/ORD-2026-05-19-Clark-County-Ordinances-5378.pdf

C3ES01 | Atlas Tube ASTM A500 rectangular HSS section properties

4x2x1/8 catalog strong I2.65/S1.33, weak I0.898/S0.898, nominal4.75lb/ft, designwall0.116in. Existing12x4x1/4 and3x1x1/8 sections retained from manufacturer table.

Limit: Current manufacturer-hosted table access is not stock, material certification, allowable stress, weld/fatigue or rail application approval.

https://www.atlastube.com/wp-content/uploads/2018/04/Atlas-HSS-Section-Properties-A500-Rectangles.pdf

C3ES02 | Steel Tube Institute HSS combined shear, axial and in-plane moment example

HSS member and local joint behavior require combined-force and wall checks; no project weld dimensions adopted.

Limit: General technical example, not the vehicle load standard or a released connection.

https://steeltubeinstitute.org/resources/hss-to-hss-connection-with-shear-axial-and-in-plane-moment-example/

C3ES03 | Retained Phase3D structural and running-gear study

D2 110member1527.287845kg benchmark,24in nodes,12inmains/bolsters and nominal596mm wheel/crossmember conflict retained as comparison inputs.

Limit: Preliminary source models, not physical measurements or OEM approval.

C3ES04 | Phase3E coordinated design basis

0%grade,28in baselinefloor,32regularplaces,13200kg upper structural case and7040kg tare allocation retained.

Limit: Owner/analyst design basis; alternative32in floor is not selected.

C3FS01 | Atlas Tube A500 rectangular HSS section properties

Retained12x4x1/4 25.82lb/ft;4x2x1/8 4.75lb/ft;3x1x1/8 3.05lb/ft table entries rechecked.

Limit: Nominal catalog properties only; no rail application, grade/toughness certification or stock confirmation.

https://www.atlastube.com/wp-content/uploads/2018/04/Atlas-HSS-Section-Properties-A500-Rectangles.pdf

C3FS02 | ASTM A500/A500M-23 public scope

Active specification page; public scope notes dynamically loaded applications may require consideration of notch toughness.

Limit: Public scope only; full material requirements and project suitability require engineer review.

https://store.astm.org/a0500_a0500m-23.html

C3FS03 | AWS D15 Committee: D15.1/D15.1M:2025

Current AWS listing describes railcar/locomotive/component welding scope for North American railroad service.

Limit: Listing is not project adoption, jurisdiction determination or approved WPS.

https://www.aws.org/About/Get-Involved/Committees/D15-Committee-on-Railroad-Welding/

C3FS04 | AWS current welding standards publication listing

D1.1/D1.1M:2025-AMD1 listed for structural steel design, fabrication, qualification and inspection.

Limit: Engineer must choose applicable vehicle welding basis; D1.1 not automatically sufficient or mandatory.

https://www.aws.org/standards-and-publications/

C3FS05 | Steel Tube Institute: welding of hollow structural sections

Explains nonuniform HSS joint force distribution and effective weld length.

Limit: Older technical article cites earlier code editions; used for engineering principle, not current weld sizing.

https://steeltubeinstitute.org/resources/welding-of-hollow-structural-sections/

C3FS06 | Steel Tube Institute: understanding HSS material specifications

Discusses material specification differences, including A1085 notch-toughness provision and weldability.

Limit: Comparison does not select grade or establish this vehicle fatigue/toughness requirements.

https://steeltubeinstitute.org/resources/understanding-hss-material-specifications-which-astm-should-i-specify-for-hss/

C3FS07 | American Galvanizers Association: tubular assembly vent/drain holes

Rectangular/square tube venting cannot be assumed from round-tube alternatives; galvanizing process requires vent/drain coordination.

Limit: No galvanizing selection or hole dimensions approved; structural/coating specialists must coordinate.

https://galvanizeit.org/knowledgebase/article/venting-and-drainage-holes-in-square-and-circular-tubular-assemblies

C3FS08 | AISC303-22 public standard listing

Search-accessible official listing identifies a buildings/bridges contract-practice framework.

Limit: Direct document access unavailable in this review; no numerical tolerance adopted; not a vehicle acceptance standard.

https://www.aisc.org/aisc/publications/current-standards/aisc-303/

C3FS09 | Clark County Ordinance5378, signed replacement Chapter22.16

Official signed ordinance provides applicability/classification, engineering, QA, inspection and commissioning provisions; adopted2026.

Limit: Building Official must determine project classification/applicability. Traveler is not approved QA or a permit; responsible Nevada professional and county interfaces remain unfulfilled.

https://www.clarkcountynv.gov/adobe/assets/urn:aaid:aem:c5347a68-d8cb-444a-9cf2-e9bf7b12080f/original/as/ORD-2026-05-19-Clark-County-Ordinances-5378.pdf

F-M01 | Voith railway couplers, Type430/530 and semi-permanent families

type430_530_published_strength_up_to_kN: 300; published_force_directions: ['tension', 'compression']; compact_folding_option: True; semi_permanent_options: ['energy absorption', 'electric heads', 'air connections']

Limit: Family maximum strength is not a project working/fatigue rating; exact variant, dimensions, angles, stroke, mass and mounting absent. No promised under-platform fit.

https://www.voith.com/corp-en/connection-components-couplings/railway-couplers.html

F-M02 | Dellner semi-permanent people-mover coupler

features: ['double-acting centre damper', 'pivot anchor/deformation unit', 'cable support']

Limit: Product-route evidence only; published page supplies no numerical rating or installation envelope. Crash integration requires whole-vehicle engineering.

https://www.dellner.com/products/semi-permanent-couplers/semi-permanent-coupler-people-mover

F-M03 | Railteco corporate RTKZ1 passenger bogie

RTKZ1_gauge_mm: 762; wheel_mm: 596; speed_kmh: 40; support: centre plate and side bearers

Limit: Not a 914.4-mm project bogie. Corporate identity corroborates manufacturer product route; no support-height drawing.

https://cn.railteco.com/cplist.aspx?cptp=8&i=14

F-M04 | Railteco RTKZ1 manufacturer product detail

axle_load_t: 3.75; tare_upper_t: 1.5; wheelbase_mm: 1165; wheel_diameter_mm: 596; gauge_mm: 762

Limit: Wrong gauge and insufficient 7.5-t two-axle total for retained 8.2947-t front-crowd reaction. Copy errors in page headings are reconciled against body/corporate identity; not a usable off-the-shelf selection. Advertised timing is not checked availability.

https://www.railwayspare-parts.com/sale-13771970-aar-standard-762m-gauge-passenger-train-bogie.html

F-M05 | Severn Lamb JUPITER exact catalog model

gauge_mm: 914; minimum_radius_m: 50; maximum_speed_kmh: 16; drive: ['diesel', 'LPG']; catalog_maximum_grade_pct: 1

Limit: No mass, adhesion, wheel/drawbar force curves, engine calibration or custom trailing-mass rating. The 30-m study is outside this published model radius; 914-mm wording requires dimensioned gauge acceptance against914.4-mm project study.

https://www.severn-lamb.com/people-movers/tracked-trains/rail-trains/jupiter-2

F-M06 | SCHOMA hydrostatic tunnel locomotive data sheet, CHL-60G

CHL60G_mass_range_t: [12, 16]; CHL60G_engine_power_approx_kW: [55, 100]; CHL60G_gauge_min_mm: 600; CHL60G_speed_approx_kmh: 25; standard_brakes: pneumatic locomotive and train; standard_coupling: Willison or similar

Limit: Industrial hydrostatic platform comparison, not passenger attraction offer. Exact914.4-mm variant, adhesive mass, radius, TE curves, emissions, styling and hydraulic brake interface need acceptance. A16-t variant exceeds14.5-t locomotive/hotel allocation before separate hotel equipment.

https://www.schoema.de/media/schoema_datenblatt_hydrostatische-tunnellokomotiven_de.pdf

F-M07 | Cummins B4.5 StageV, B4.5 173 calibration row

calibration: B4.5 173; rating: Intermittent; rated_kW: 129; rated_rpm: 2000; peak_torque_Nm: 780; peak_torque_rpm: 1500

Limit: Intermittent power is not continuous site output. No chosen CPL, allowable duty definition, cooling rejection, fuel curve, aftertreatment package or US project application approval. Peak torque is not available by assumption at2000rpm.

https://www.cummins.com/en-apac/engines/products/b45-stage-v

F-M08 | Irwin locomotive tractive-effort chart

chart_10_ton_column_TE_lbf: 5000; chart_15_ton_column_TE_lbf: 7500; chart_20_ton_column_TE_lbf: 10000

Limit: Generic manufacturer chart, not exact model or speed-dependent rating. Do not use catalog gross/train tonnage as project capacity. Ton-unit convention and rating conditions require confirmation; force conversion alone is independent of those units.

https://www.irwincar.com/locomotive-tractive-chart/

F-M09 | Irwin diesel/hydraulic industrial locomotive family

published_upper_size_tons: 40; transmission_up_to15tons: single-speed constant mesh; larger_units: multi-speed

Limit: No exact project model,914.4-mm axle arrangement, drawbar curve, coach brake integration or confirmed availability.

https://www.irwincar.com/rail-products/locomotive-industrial/

F-M10 | SCHOMA narrow-gauge locomotive capability

application_routes: ['tourist railways', 'work and field railways']; custom_transport: True

Limit: Capability only, not a quoted passenger locomotive configuration.

https://www.schoema.de/de/lokomotiven/produkte/schmalspurlokomotiven/

C3FB01 | IFE - STR and X 4 sliding steps

STR: 900-1800 mm clear width, 0-600 mm extension, 56 mm installation height. Published example is 65 kg at 1300 mm cassette/360 mm extension.

Limit: Single sliding boarding aid, not a complete 32 in ground staircase. Published example mass is not this project mass; load rating and configured CAD are absent.

https://www.ife-doors.com/en/products/access-devices/sliding-step/

C3FB02 | BODE - BCBS step systems

Customized single/multistage kinematics. Published entry range 800-1500 mm; locking switch and load/platform detection options. Example 68 kg at 1300 mm width/270 mm stroke.

Limit: Configured clearances, full ground-stair capability, loads and controls require proposal. Product-family claims do not approve this installation.

https://www.bode-global.com/en/products/boarding-aids/bcbs

C3FB03 | Wabtec - Access sliding step brochure

2018 brochure: example 1300 mm door, 57 mm thickness, 65 kg, 150-500 mm stroke; integrated door/step control and mechanical holding features described.

Limit: Historical reference hosted by current manufacturer. Not a current quote or full flight of stairs; exact application acceptance required.

https://www.wabteccorp.com/media/5626/download?inline=

C3FB04 | BraunAbility - NCL 1000 application guide, 3 June 2026

NCL1000IB3754HB-2: 385 lb mass, 1000 lb capacity, 37 × 54 in clear platform, 48 in travel, 50.96 in overall width, 45.85 in minimum door width, 67 in minimum door height, 18 in threshold area; 12 VDC/70 A maximum.

Limit: Motor-vehicle product; no rail approval or actual mounting/swept CAD. Guide reduces clear vertical passage by 1.375 in from door opening. No actual configuration selected.

https://www.braunability.com/content/dam/braunability/docs/comm/na/app-guide/updated-dpa-lift-app-guides/Lifts-Comm-Century-1000-Guide-6-3-26.pdf

C3FB05 | Stepless/Guldmann - LP 11 technical data, October 2024

LP11 model 13011: 300 kg capacity, 180-955 mm lift range, 149 kg mass, 850 × 1400 mm platform, approximately 1140 × 1590 mm overall; upper ramp 610-792 mm wide and 1000 mm long.

Limit: Use technical maximum 955 mm instead of rounded website 1 m. Actual deployed CAD and US/project acceptance needed. Published electrical data need configuration confirmation.

https://media.guldmann.com/cdn/5FPGwx4/datasheet-lp11-gb.pdf

C3FB06 | Stepless/Guldmann - LP 11 operating manual, July 2024

Manufacturer requires trained operation, stationary train, even stable support and applied lift brakes. Manual addresses ramps, lowering and transport.

Limit: Not an all-terrain rescue solution. Current configuration and complete manual govern; no occupied demonstration authorized by this report.

https://media.guldmann.com/cdn/5FPGwx3/manual-lp11-gb.pdf

C3FB07 | Adaptive Engineering - Mobilift TX current product page

Station-based manually powered rail lift with 600 lb capacity; product page gives approximately 268 lb unit mass.

Limit: Mass conflicts with linked brochure 295 lb and approximate 298 lb wording. Marketing statements do not replace actual route/slope and installed interface acceptance.

https://www.adaptivelifts.com/tx.php

C3FB08 | Adaptive Engineering - Mobilift TX technical brochure

60 in maximum lift; 600 lb capacity; approximate 37.5 ×72 in overall plan; 34 ×56 in platform; standard bridge 20 ×29 in and ground ramp 30 ×49 in. Movement while occupied is prohibited.

Limit: Brochure has mass and metric-conversion inconsistencies. Published 29 in bridge needs reconciliation with applicable 30 in width rule. OEM must provide exact configured dimensions/ratings.

https://www.adaptivelifts.com/pdf/Mobilift_TX_Brochure.pdf

C3FB09 | US Access Board - amusement ride technical requirements

Where applicable, §1002 addresses wheelchair maneuvering, companions and loaded station gaps; normal-loading floor offsets are limited or require appropriate boarding devices.

Limit: The 42 in opening/60 ×36 in side-entry example is not approval. Full scope, exceptions and actual equipment must be reviewed.

https://www.access-board.gov/ada/chapter/ch10/

C3FB10 | US Access Board - vehicle mobility aid accessibility

§1192.83 addresses lifts and ramps/bridges: ratings, retention, stowage, barrier, slope and operation provisions. Amusement gap exceptions reference applicable ramp/bridge portions.

Limit: Does not automatically classify this railway as covered light rail. Regulatory minimum loads do not establish the required actual occupant/device rating.

https://www.access-board.gov/ada/vehicles/subpart-d/

C3FB11 | US Access Board - vertical clearance §307.4

General 80 in vertical clearance provision has a 78 in exception for door closers/stops.

Limit: Not a blanket 78 in structural doorway allowance. Applicability and vehicle-specific provisions require review.

https://www.access-board.gov/ada/chapter/ch03/

C3FB12 | US Access Board - stairs and handrails §§504-505

Where scoped, uniform 4-7 in risers, at least 11 in treads, closed risers, wet-condition drainage and two-sided handrails are addressed.

Limit: Used as a conditional planning benchmark, not an adopted vehicle stair standard. Guards and attachments still require an established basis.

https://www.access-board.gov/ada/chapter/ch05/

C3FB13 | US Access Board - routes, ramps and facility platform lifts

Where applicable, §405 limits ramp-run rise to 30 in and addresses landings; §410 facility lifts differ from vehicle lifts.

Limit: Facility/vehicle classification, ramp branches, headroom and maneuvering requirements need coordination. No facility approval inferred.

https://www.access-board.gov/ada/chapter/ch04/

C3FB14 | Clark County - signed Ordinance 5378, effective 3 June 2026

Replaced Chapter 22.16. Scope includes installations 30 days or longer subject to exceptions; Building Official classification and Nevada design-professional engineering apply as scoped.

Limit: See root jurisdiction update. Exact project classification and adopted referenced-standard editions remain unresolved; operating evenings do not define installed duration.

https://www.clarkcountynv.gov/adobe/assets/urn:aaid:aem:c5347a68-d8cb-444a-9cf2-e9bf7b12080f/original/as/ORD-2026-05-19-Clark-County-Ordinances-5378.pdf

C3FB15 | US Access Board - amusement ride classification guide

Distinguishes amusement ride use from primarily transporting passengers; circulation/load-unload headroom differs from the wheelchair ride-space exception.

Limit: North Pole disembarking is relevant but does not independently settle classification.

https://www.access-board.gov/ada/guides/chapter-10-amusement-rides/

C3FU01 | FreshJet mechanical ADB installation and operating manual

FJX3573M: 15, 000 Btu/h, 115 V, 60 Hz, single phase, compressor 13.6/71 A rated/locked, fan 2.6/6.9 A. FJX5557M: 15, 000 Btu/h, 12.8/57 A compressor, 3.6/8.2 A fan. FreshJet 3 body 29.6 × 27.6 × 14.4 in; FreshJet 5 body 40.39 × 29.21 × 11.1 in. RV application and roof/framing limits require vehicle review.

Limit: No rail application acceptance, 110°F delivered capacity, PF or project starting duration. Mechanical air-distribution box intrudes below the roof; no retained-headroom installation is established. Manual conductor/protection examples are not adopted as train design.

https://media.dometic.com/externalassets/dometic-freshjet-3-series-15k_107740.pdf

C3FU02 | NW-4 rooftop HVAC

Rail rooftop family normally 480 V three phase; other voltages possible by configuration; up to 11 tons cooling and 24 kW heat, staged output.

Limit: Family maxima are not a sized coach selection. Configured dimensions, mass, starting input and hot-site capacity absent.

https://nwrail.com/nwre_products/nw4.html

C3FU03 | NW-2 undercar HVAC

Rail undercar package with staged cooling/heating and configured condensate drainage.

Limit: No claim that this full-size package fits the F1 seven-inch service reservation. No current project offer.

https://nwrail.com/nwre_products/nw2.html

C3FU04 | Spring-applied actuators

Mechanical springs supply brake application energy after energy loss; families may include a second hydraulic release circuit and auxiliary release.

Limit: No selected force convention, service modulation map, disc compatibility or single-fault acceptance.

https://www.hanning-kahl.com/products/hydraulic-brake-systems/spring-actuators.html

C3FU05 | HCU hydraulic unit

Configured modular hydraulic units include serviceability and filtration features; function depends on configured components.

Limit: No current, pressure, flow, size or package rating is assigned from this page.

https://www.hanning-kahl.com/products/brakemaster/hydraulic-unit-of-the-hcu-series.html

C3FU06 | Auxiliary release devices

Auxiliary pressure can release spring brakes through a second hydraulic circuit independently of service release.

Limit: This is a recovery-release capability, not independent brake application. Trapped pressure/isolation must be addressed.

https://www.hanning-kahl.com/products/hydraulic-brake-systems/auxiliary-release-devices.html

C3FU07 | Safety Guide 4400-B.1

Assembly selection must account for pressure/transients, temperature, fluid compatibility, bend radius, motion/twist, protection, cleanliness and inspection.

Limit: General manufacturer guidance; selected brake hose/fitting and rail application need OEM acceptance. No generic hose size or replacement interval adopted.

https://www.parker.com/content/dam/Parker-com/Literature/Parflex/Parker-Safety-Guide.pdf

C3FU08 | Rail hydraulic brake systems

Rail product range includes active and spring-applied constructions and electro-hydraulic units.

Limit: Shows architectural alternatives; no mixed-supplier package is qualified.

https://www.knorr-bremse.us/en/products/braking-systems/hydraulics/